

Question Paper

Exam Date & Time: 22-Dec-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE MAKEUP EXAMINATIONS - DECEMBER 2025

Text Analytics [CSE_4006]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) What is information extraction? Explain how does it bridge the gap between unstructured text and structured data? (5)
- A) Course Outcome: "CO1" / Cognitive Level: "Understand"
- B) Contrast basic text mining and information extraction approaches. (5)
- Course Outcome: "CO1" / Cognitive Level: "Analysis"
- 2) Identify the differences between stemming and Lemmatization techniques using suitable examples. (5)
- A) Given a large-scale real-time search-system with millions of documents and speed is critical but exact word-meaning matching is less critical, suggest suitable method would you choose stemming or lemmatization with reasons. (5)
- Course Outcome: "CO2" / Cognitive Level: "Application"
- B) Identify the algorithmic steps for generating Multiword features from tokens using suitable example. (5)
- Course Outcome: "CO2" / Cognitive Level: "Application"
- 3) A global business intelligence firm is developing a Natural Language Processing (NLP) system to analyze news articles and company reports for identifying investment opportunities. The system needs to extract meaningful insights from sentences such as: (5)
- A) *"Apple Inc. acquired an AI startup in San Francisco for \$200 million"*
Apply NLP techniques of Entity Recognition, Phrase Recognition, Parsing, in this scenario to analyze the sentence. Describe how each step contributes to extracting useful business insights and improving the performance of downstream machine learning models. (5)
- Course Outcome: "CO3" / Cognitive Level: "Application"
- B) Experiment the working of the Sentence Boundary Determination algorithm. Apply the Sentence Boundary Determination algorithm to the following text and identify the correct sentence boundaries. (5)
- "Dr. Kumar teaches AI. He lives in Chennai! Isn't that wonderful?"*
- Course Outcome: "CO3" / Cognitive Level: "Application"
- 4) A Natural Language Processing (NLP) researcher is studying word associations in a large collection of technology news articles. In the corpus of 10,000 sentences: (5)
- A) The word "AI" appears in 600 sentences

The word "robotics" appears in 400 sentences
Both "AI" and "robotics" appear together in 150 sentences

1. Identify Pointwise Mutual Information (PMI) between the words "AI" and "robotics."
2. Analyze what the computed PMI value indicates about their relationship in the corpus.

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- B) A biomedical research organization is developing an NLP system to analyze relationships between medical terms in research articles using a domain ontology such as MeSH or SNOMED CT. (5)
- For instance, the ontology defines hierarchical relationships like:
Disease → *Viral Disease* → *Respiratory Infection* → *COVID-19*
Disease → *Viral Disease* → *Respiratory Infection* → *Influenza*
- Suppose in our corpus the estimated probabilities are:
 $P(\text{COVID-19})=0.01$
 $P(\text{Influenza})=0.02$
 $P(\text{Respiratory Infection})=0.05$
1. Solve both Resnik and Lin similarity values for *COVID-19* and *Influenza*.
 2. Explain which similarity measure is more suitable for comparing concept pairs across an ontology, and why.

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5) Compare how machine learning automates entity recognition with handcrafted rule-based systems. (5)
- A) Categorize the entities and labels found in the sentence "Robert L. James is the CEO of McCann-Erickson."

Course Outcome: "CO5" / Cognitive Level: "Analysis"

- B) Model the template-filling process for the sentence "Dr. Jon Smith from IBM will present a seminar on Tuesday," and solve the problem of missing slot values through appropriate techniques. (5)

Course Outcome: "CO5" / Cognitive Level: "Application"

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Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026 CSE_4006:Text Analytics

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026 CSE_4006:Text Analytics

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	1	5(10%)

3	Application	6	30(60%)
4	Analysis	3	15(30%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
	Total	10	50 (100%)